reach already runs end to end for Tübingen; we show you the app live

1,699

populated 100 m cells (90,355 residents)

15 × 4

categories × profiles (walking at two speeds, cycling, wheelchair)

9 of 11

validation experiments done

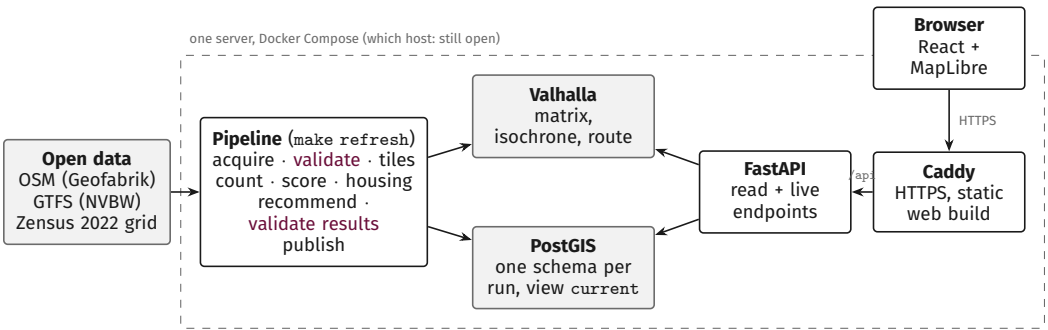
What we show in the app

1. **Map:** who lacks access to what, per category and profile
2. **Click anywhere:** what is reachable in 15 minutes, with times and routes, and the nearest missing facility
3. **Combined score** 0–100 over the categories you pick
4. **Recommendations:** where a new facility would help the most residents (224 zones)
5. **Housing potential:** empty cells with good access for new homes (830 candidates)
6. **Data page:** sources, snapshot dates, validation report

Runs locally today; pipeline, API and web app are on `dev`.

Source: run 20261110T233930Z (OSM 8 Nov 2026); validation report (run 20261106T142630Z); docs/features.md §1–§5.

How it works: one command turns open data into the map



Open data only: OSM, timetables, census grid; one snapshot per run.

Real travel times: routing on the street network, not straight lines.

Checked: 15 automatic checks; a failure keeps the last good run.

Source: docs/architecture.md §2, §3, §7, §8; run 20261110T233930Z (validate_results report); weekly refresh and production server are planned (§11).

We tested our method against the obvious alternatives, and it holds up

+18 %
wrong pairs if we counted with isochrone polygons
Mostly across the Neckar, rail lines and main roads. We count with real travel times instead.

≤ 2.8 %
below the optimum: our fast recommender vs. an exact solver
90 test instances; good enough to run in every refresh.

88 %
of the city's Kitas found in OSM
Schools: 95 %. The gaps are known; we can map them in OSM.

1.9 %
change class if we used buildings instead of cell centres
Walking; the 100 m grid is precise enough.

Still to do: a check against local knowledge and a full reproducibility run. All results: backup.

Source: validation report V1, V2, V4, V6 (run 20261106T142630Z); walking 3.39 km/h, 15 min.

Tübingen fits on one small server; all of Germany would need a bigger setup

Server for Tübingen

- Counting all 15 categories $\times$ 4 profiles takes 254 s on a laptop
- The routing engine needs up to 4.0 GiB of memory, so we plan for 8 GB RAM
- **Option A:** rented VPS ($\geq$ 8 GB); budget 20 EUR in total, so online from about 1 Dec until grading (price still to check)
- **Option B:** our own HP ProDesk (i5, 16 GB): no cost, but home network and uptime
- Optional: web frontend separately on Cloudflare Pages (proposal)

All of Germany (idea, not planned)

- The census file we already use covers 3,088,037 populated cells (82.6 million residents), about 1,800$\times$ Tübingen
- Same grid, same pipeline, one run per municipality
- Heavy routing builds on bwUniCluster instead of our server
- Hard parts: OSM quality differs by region; villages need other thresholds than a city

Source: docs/architecture.md §11, §11.1; docs/deployment_proposal.md (PR #64, not approved); notes N12, N16 (laptop, run 20261110T233930Z); Zensus file of run 20261110T233930Z (grid sum).

What we would like to discuss with you today

Question	Our suggestion
Server: Is our own ProDesk fine, or do you expect a rented VPS? How long does the app need to stay online?	Measure one full refresh on the ProDesk, then decide
Access: How would you like to look at the data: app only, or also the database (e.g. via SSH tunnel)?	Read-only database account
Scaling: Is Tübingen enough, or would a second city or a larger region be interesting? May we use bwUniCluster?	Finish Tübingen first; Germany as outlook
Method: Moran's I shows strong clusters on cells (0.72–0.88). Is that enough as the "adequate method", or do you expect more?	Optimisation plus spatial statistics
Limits: Is it okay to name the known limits openly (wheelchair on slopes, OSM gaps, new categories not validated yet)?	Yes, on the data page and in the report
Interim on 25 Nov: What do you expect: live demo, results, or both?	Live demo with backup screenshots

Source: docs/deployment_proposal.md (PR #64); methodology.md §7 (M5); validation report V8 (run 20261106T142630Z); notes N11, N13, N16; project_management_decisions.md §8.

Next steps: server live in December, submission on 22 January

Plan until the submission

- **25 Nov:** interim presentation (`v0.1.0`)
- **from ∼1 Dec:** server live, weekly automatic refresh
- **Dec–Jan:** remaining validation (local knowledge, reproducibility); final categories and public-transport classes
- **10–15 Jan:** feature freeze
- **22 Jan:** submission (`v1.0.0`); **29 Jan:** final presentation

What we need from today

- A decision on the server, so we can deploy from 1 Dec
- Your view on scope: Tübingen only, or a look beyond
- Feedback on the app: what is missing, what is too much?

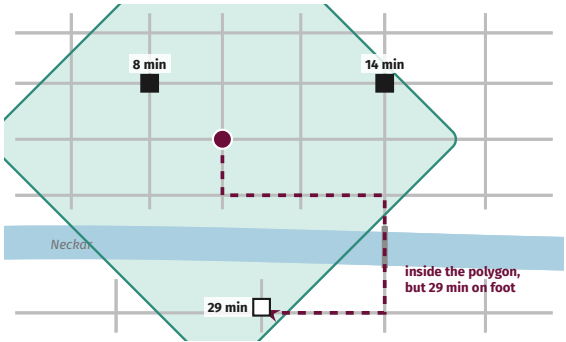
Source: docs/features.md §9; architecture.md §11; methodology.md M1, M2.

Backup: validation experiments V1–V11

#	Experiment	Key result
V1	Matrix vs. isochrone polygons	Polygons add 18–20 % false pairs, 47 % across barriers
V2	OSM completeness	Childcare recall 88 %, precision 98 %; schools 95 % / 95 %
V3	Recommender back-test	80–98 % of picks within 500 m of a removed facility (only the removed deficit counts)
V4	Greedy vs. exact MILP	At most 2.8 % below the optimum (mean 0.17 %)
V5	Sensitivity (k, speed, budget)	Shares move strongly; district rankings stable (median Spearman 0.78–0.95)
V6	Centroid vs. building origin	1.9 % (walking) / 2.8 % (wheelchair) change class
V7	Local knowledge	Not yet run
V8	Spatial statistics	Moran's I 0.72–0.88 on cells; none between districts
V9	Reproducibility	Not yet run
V10	Matrix vs. single routes	Median $\lvert\Delta\rvert$ 0.5 s; 0 of 900 false negatives
V11	With vs. without slopes	3.3–3.6 % change class, mostly on hillsides

Source: docs/validation_report.md, run 20261106T142630Z (Core categories of that run, parks and playgrounds still one category); the extended categories are not validated yet.

Backup: counting with isochrone polygons would add 18 % unreachable pairs



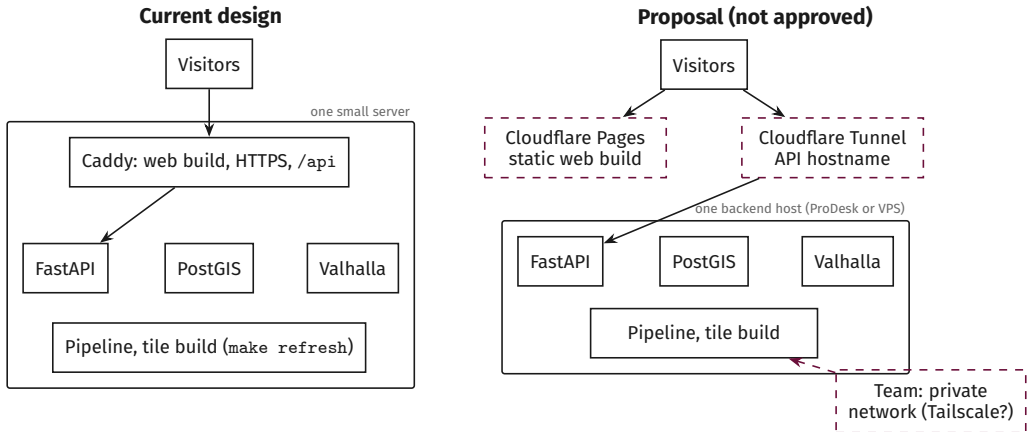
How we count: per block of cells one travel-time matrix to all facilities in reach; a facility counts if its time is $\leq$ 15 min. Isochrones are only drawn.

Test (150 cells, walking 3.39 km/h): polygons add 18 % unreachable pairs, 47 % of them across the Neckar, rail lines or main roads.

Nothing lost: 0 of 1,200 sampled pairs outside the counts are reachable by a single route.

Source: docs/architecture.md §6.3, §6.5; validation report V1 (run 20261106T142630Z); run 20261110T233930Z (`false_negatives`).

Backup: hosting, current design vs. the Cloudflare proposal



Source: docs/architecture.md §3, §11 (current); docs/deployment_proposal.md (proposal, PR #64): backend on the HP ProDesk (i5-8500T, 16 GB) or a rented VPS ($\geq$ 8 GB).

Backup: known limits and how we handle them

Limit	Effect	Proposal
Wheelchair on slopes	Steep sections are not avoided; hillside counts are optimistic	Accept, state on data page
OSM gaps	13 of 104 Kitas and 2 of 38 schools have no OSM match	Map them in OSM
New categories	No completeness check yet; consistency check at 10 of 1,699 cells, fails above 1 %	Mark as indicative
Cycling recommender off	Would add only 12 zones but needs 4.0 GiB of routing memory	Accept, document
Housing candidates	Land use not fully excluded; buildable not known	Decide after map check

Shares depend strongly on speed and budget: we always name the profile and budget next to a share.

Source: notes N11, N12, N13, N15, N16; validation report V2, V5 (run 20261106T142630Z); run 20261110T233930Z.

Backup: we use the official census 100 m grid instead of H3 hexagons

	Census 100 m grid (chosen)	H3 hexagons
Residents	Exact counts as published	Re-gridding spreads residents by area; errors at every edge
Identity	Official cell id, joins other census tables (e.g. age)	Own index; every join needs re-gridding
Scale	Same grid for all of Germany: 3,088,037 populated cells in the file we already download	Global, but residents must be disaggregated first
Trade-off	Squares: 8 neighbours at two distances	Hexagons: 6 equal neighbours, nicer for spatial statistics

Source: docs/architecture.md §2, §5.2, §12; validation report V6 (run 20261106T142630Z); Zensus file of run 20261110T233930Z.